PAIR: an appeal copilot that shows its sources

Helping Medicare patients fight prior authorization denials. The AI reads; it never decides.

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Try the live demo

Code on GitHub

3.6 million — **Medicare Advantage denials in 2024 were never appealed**
4.1 million denied, 88.5% not appealed · KFF, 2024 data

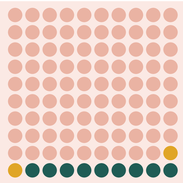
→ **80.7%** — **of appealed denials win, but only 11.5% get appealed**
Medicare Advantage · KFF, 2024 data

→ **99.4%** — **of sentences in PAIR's letters are backed by a quoted source**
153 of 154 kept sentences · 30 test cases

1 The problem

9 in 10 denials are never appealed, though most appeals win

Medicare Advantage plans cover about half of people on Medicare. Many treatments need the plan's OK first (prior authorization). In 2024 these plans denied **4.1 million requests**. Most appeals win, yet **only about 1 in 9 denials is appealed**. To appeal, a patient has to match dense coverage rules, line by line, to their own medical records.



Out of every 100 denials

- **89** never appealed
- **2** appealed, still denied
- **9** appealed and won

Of every 100 denials, about 11 are appealed and 9 of those win (KFF, 2024 data).

Why now: rule CMS-4201-F (2024) makes these plans follow Medicare's own written coverage rules (NCDs and LCDs), so an appeal that quotes them carries weight.

2 What PAIR does

Upload a denial and records; get a checked, sourced appeal

Most AI in prior authorization helps insurers decide faster. **PAIR works for the other side: it helps patients answer back.** A patient uploads the denial letter and their medical records. PAIR then:

Understand the denial

- **Explains** it in plain words: what was denied and why.
- **Checks** each Medicare coverage rule against the records: met, not met, or not sure.
- **Finds the gaps:** lists missing documents and who can provide them.

Answer back

- **Drafts the appeal letter:** every sentence quotes its source in the records or the rule.
- **Estimates the chance of winning** honestly, and gives no number when the data is too thin.

Working today: a live web demo with 7 screens, Word and PDF export, and **1,367 automated tests**.

3 What's new, built safe

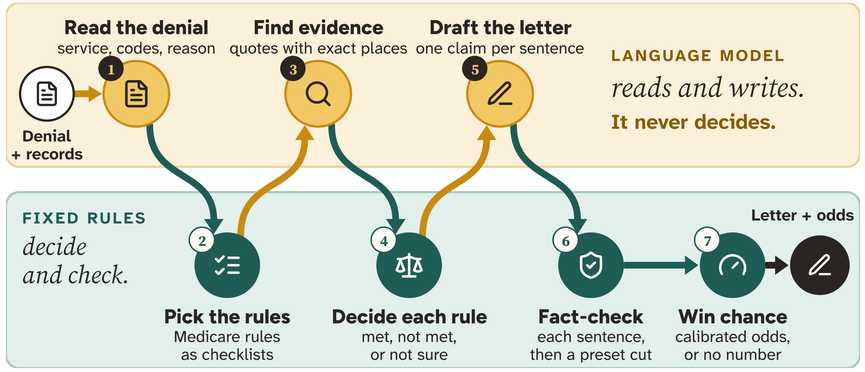
What a general chatbot can't do

	For the patient	Each sentence sourced	Checks the Medicare rule	Honest odds
General AI chatbot	✓	✗	✗	✗
Insurer-side AI tools	✗	*not public*	*not public*	*not public*
PAIR	✓	✓	✓	✓

Safe by design: made-up patients only, uploads never logged, models run on our own machines, and when unsure PAIR says **"talk to a person first"** (free Medicare counselors).

4 How PAIR works: the AI reads, fixed rules decide

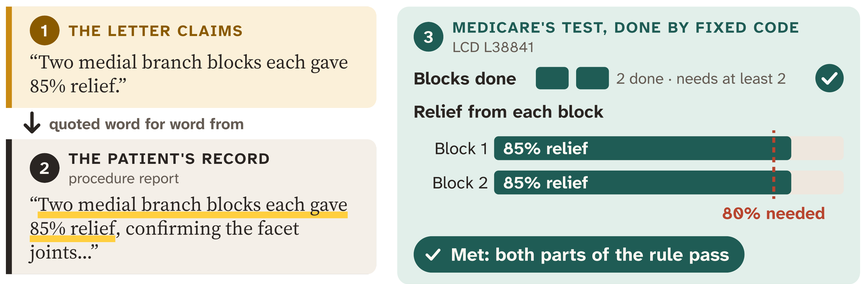
The language model only reads and writes; fixed rules make every decision

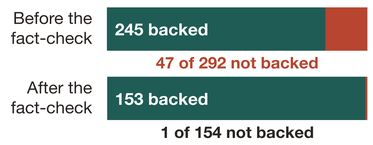


A language model can misread or make things up, so in PAIR it never decides.

5 Every sentence is checked before you see it

1 in 6 draft sentences lacked proof; after the check, 1 of 154



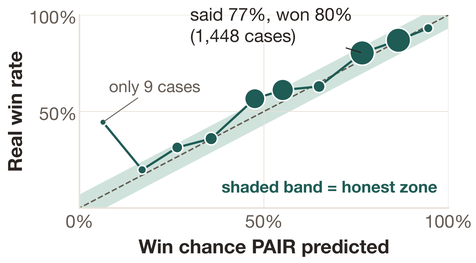


Before fact-check, **1 in 6** drafted sentences was not fully backed by the records. After it, **1 of 154**.

We set the limit before testing: at most 15% of bad sentences may slip through. Actual: 1 of 47.

6 Odds you can trust

Its win chances match real outcomes on 5,991 unseen cases



When PAIR says 70%, about 70 in 100 such cases really won (5,991 past reviews it never saw).

Bigger dots hold more cases. 8 of 10 groups sit in the shaded honest zone, within 7 points of the truth.

7 Results: every target met

Every target met, each set before testing

What we measured	Result	Target	Dev
Reads the denial right	**100%**	≥ 90%	100%
Decides each rule (F1*)	**0.90**	≥ 0.85	0.94
Facts traced to the record	**99.5%**	≥ 95%	99.2%
Unsupported sentences let through	**1 of 47**	≤ 15%	
Win ranges that hold	**91.7%**	90%	
Upload to cited letter	**15.3 s**	< 60 s	

*Dev: 30 cases on other made-up patients. *F1: match with the answer key (1.0 is perfect).*

- **How we tested:** 30 test and 30 held-out cases on separate made-up patients. **Pass rules written down before each run.**
- An audit planted 20 known errors in the cases; an independent checker **caught all 20**.
- **Win chance tested on the future:** the model learned from older cases (up to 2023) and was tested once on 5,991 newer ones (2024 to early 2026) it had never seen.

8 Limitations

Promising on made-up patients; not yet proven on real records

- **Small rulebook so far:** 3 treatments, built by hand from 5 public Medicare rule documents. Medicare has hundreds more, and private insurers write their own.
- Win chance learned from California commercial plan reviews, not Medicare: **a base rate, not a promise**.
- Not legal or medical advice; a person should read every letter.

9 Impact: who gains

3.6 million unappealed denials a year is the opportunity

- **Patients:** PAIR turns pages of dense rules into a checked draft in about 15 seconds, which the patient reviews and sends.
- **Helpers:** free Medicare counselors (SHIP) and clinic staff could prepare more appeals in less time, with every sentence linked to proof they can check.
- **Beyond Medicare:** the same pattern (AI reads, fixed rules decide, every sentence sourced) fits other high-stakes paperwork, such as disability or benefit denials.
- **What's next:** grow the rulebook (more Medicare rules first, then private insurers' published policies, with a model drafting each checklist and a person checking it), a view for clinic staff, and a pilot on real records with patient consent.

Illustration, not a result: if just 1 in 100 unappealed denials were appealed and won at today's rate, about **29,000 more people** would get care.

Built with

Papers Mohri & Hashimoto 2024, conformal factuality (ICML) · Rehman & Yeh 2026, conformal factuality for multi-hop RAG (arXiv 2609.38222) · Zhang et al. 2026, Verifiable by Construction (arXiv 2609.15964) · Lodhiya et al. 2026, AGVF (arXiv 2609.27844) · Tang, Laban & Durrett 2024, MiniCheck (EMNLP) · Vovk & Petej 2014, Venn-Abers predictors

Tools Gemma 4 E4B (Ollama), Bespoke-MiniCheck-7B, Gemini 3.5 Flash (wrote test cases), LangExtract, Docling, scikit-learn, venn-abers, Presidio, FastAPI, Next.js. Coding help: Claude. Icons: Lucide.

Data CMS Medicare Coverage Database (5 rule documents), California DMHC independent medical reviews (42,749), Synthea (made-up patients), KFF report (Jan 2026).